

Ground floor study

Where the garage stair lands, and the floor planned around it: a tall living room behind a pentagon of glass, the stair up at the back beside the toilet, the house a metre deeper into the hill. Metres, to scale, house frame: x along the house (the valley at the bottom), z across (north to the right).

9.00 × 7.00

over all, a metre further into the hill; 8.40 × 6.40 clear, 53.8 m²

x 5.70

where the garage flight lands: the back wall line, 2.30 of landing to the stair wall

x 2.3–3.3

the stair up: one flight along the back wall, 17 × 17.1 on 0.27 treads, beside the toilet

6.40 × 9.10

the panorama: one pentagon of glass, 5.90 at the walls, 9.10 at the apex, the living room's alone

45°

the roof: eaves at 5.60 on the walls, ridge at 9.10; the living room is open to it

22.6 · 10.2 · 7.8 · 2.2

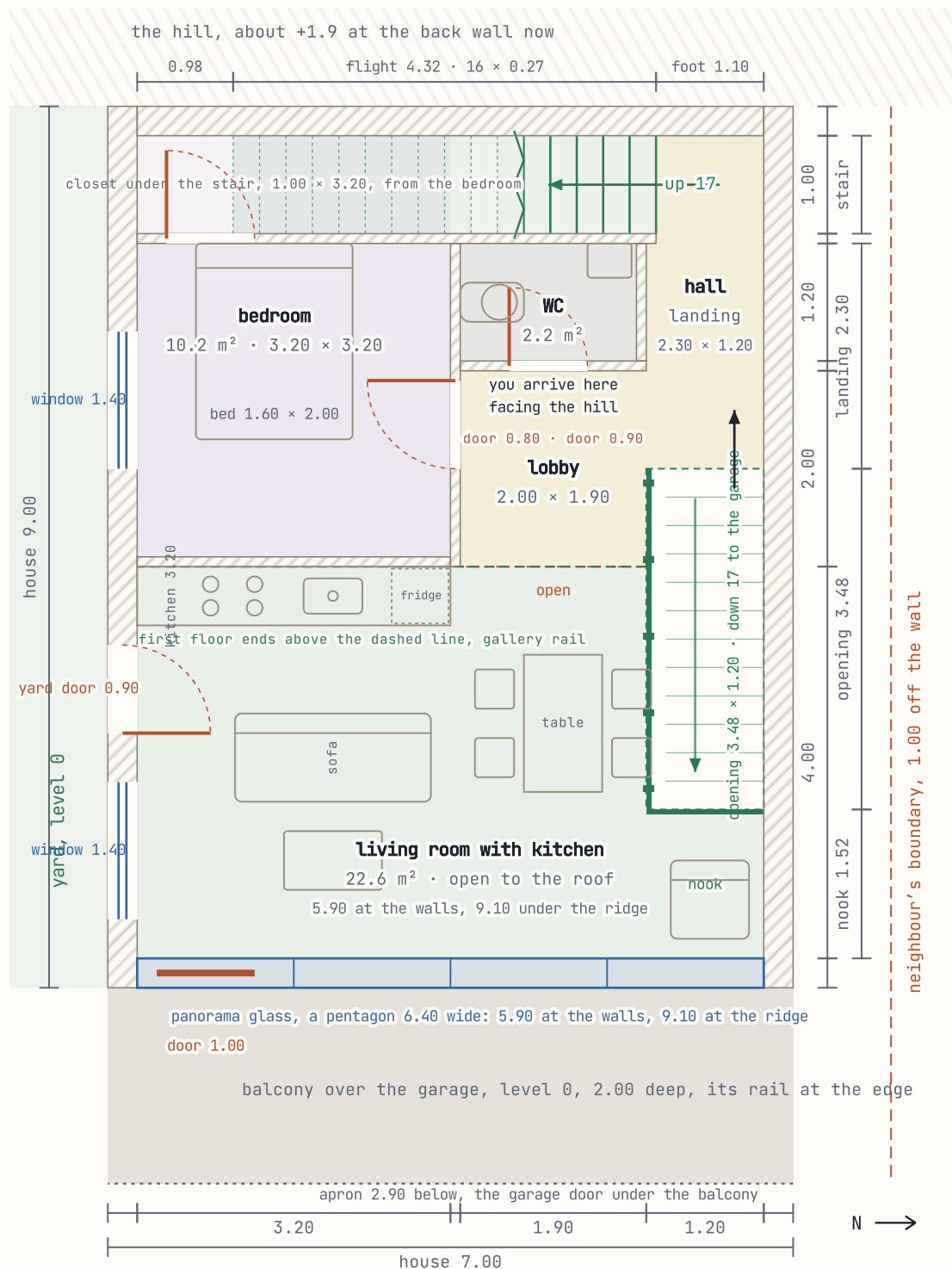
living room with kitchen, bedroom, hall, toilet, m²; and a 3.2 closet under the stair

24.9

first floor behind the tall room, m² clear: 4.40 × 6.40 less the stairwell

Plan, 1 m above the floor

The hill is at the top, the valley and the balcony at the bottom, the yard to the left (south), the neighbour's boundary 1 m off the right (north) wall. Walls 0.30, partitions 0.10.



- hall
- toilet
- bedroom and its closet under the stair
- living room with kitchen, open to the roof
- glass

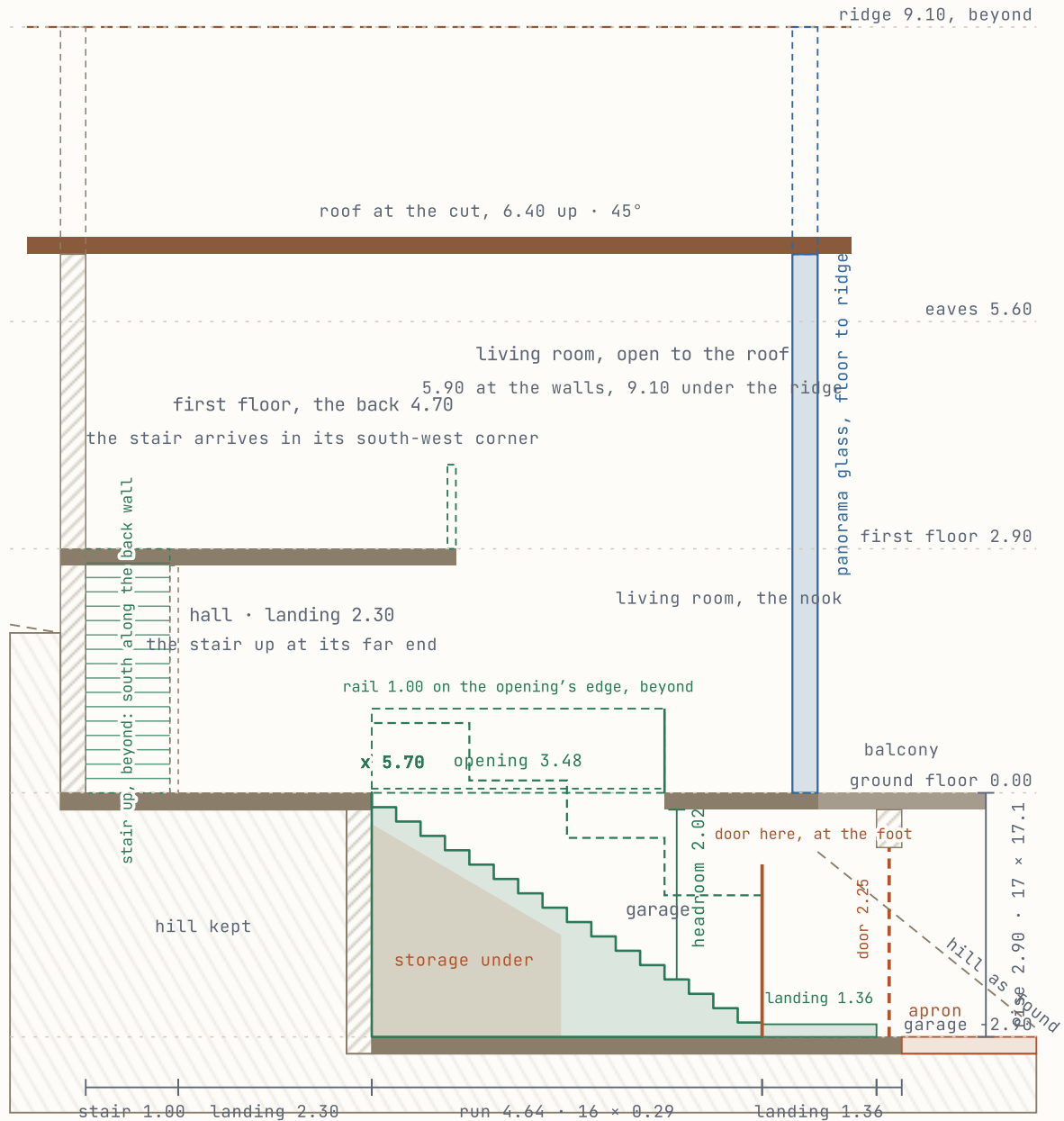
- **The hall is where the stair lands.** You come up facing the hill at x 5.7, with 2.30 of landing to the stair wall; at the landing's west end the stair up turns south along the back wall. The lobby south of the

landing has the toilet door on its west side and the bedroom door on its south side, and opens east into the living room.

- **The stair up** is one straight flight along the back wall, $\times 2.3\text{--}3.3$, beside the toilet: 17 risers of 17.1 on 0.27 treads, foot at $z\ 1.4$, top riser at $z\ 5.72$, 0.98 of landing to the south wall at the top, so it arrives at the first floor's south-west corner with the bathroom over the toilet beside it. A wall on its east side is the toilet's and the bedroom's west wall; under its upper half is the bedroom's closet, 1.00×3.20 , 1.2 to 2.7 high, with its door at $z\ 5.5$.
- **The living room is the whole front**, 4.00×5.20 plus the nook by the garage opening, and it is tall: the first floor stops at $x\ 6.7$, on the bedroom wall, so from the kitchen to the glass the room rises past the first floor to the roof — 5.90 at the walls, 9.10 under the ridge. Its east wall is the panorama, and nothing else stands behind it: one pentagon of glass, 6.40 wide, from the floor to the roof, with a 1.00 door to the balcony at its south end. The garage opening is railed on its south edge and at its east end.
- **The entrance from the yard** is a 0.90 glazed door in the south wall beside the kitchen; a 1.40 window further along lights the sofa.
- **The kitchen** stands against the bedroom wall, 3.20 of counter with the fridge at the hall end. The table sits by the garage opening, the sofa faces the glass, an armchair in the nook.
- **The bedroom** is 3.20×3.20 in the south-west corner with a 1.40 window to the yard: a 1.60×2.00 bed with its head to the stair wall, and the closet under the stair instead of a wardrobe. Its west side and the toilet's are the stair; the back wall beyond it is against the hill.
- **The first floor keeps the back 4.70 m** ($\times 2\text{--}6.7$): 4.40×6.40 clear less the stairwell, 24.9 m^2 , with a gallery rail on its edge over the living room and the stair arriving in its south-west corner.

Where the garage stair lands: section along it, looking south

Cut 0.8 m from the north wall, through the garage flight. Garage floor -2.90, ground floor 0.00, first floor 2.90, eaves 5.60, ridge 9.10. The roof is cut where it is 6.40 up; the ridge is beyond, and so is the stair up, seen rising along the back wall.

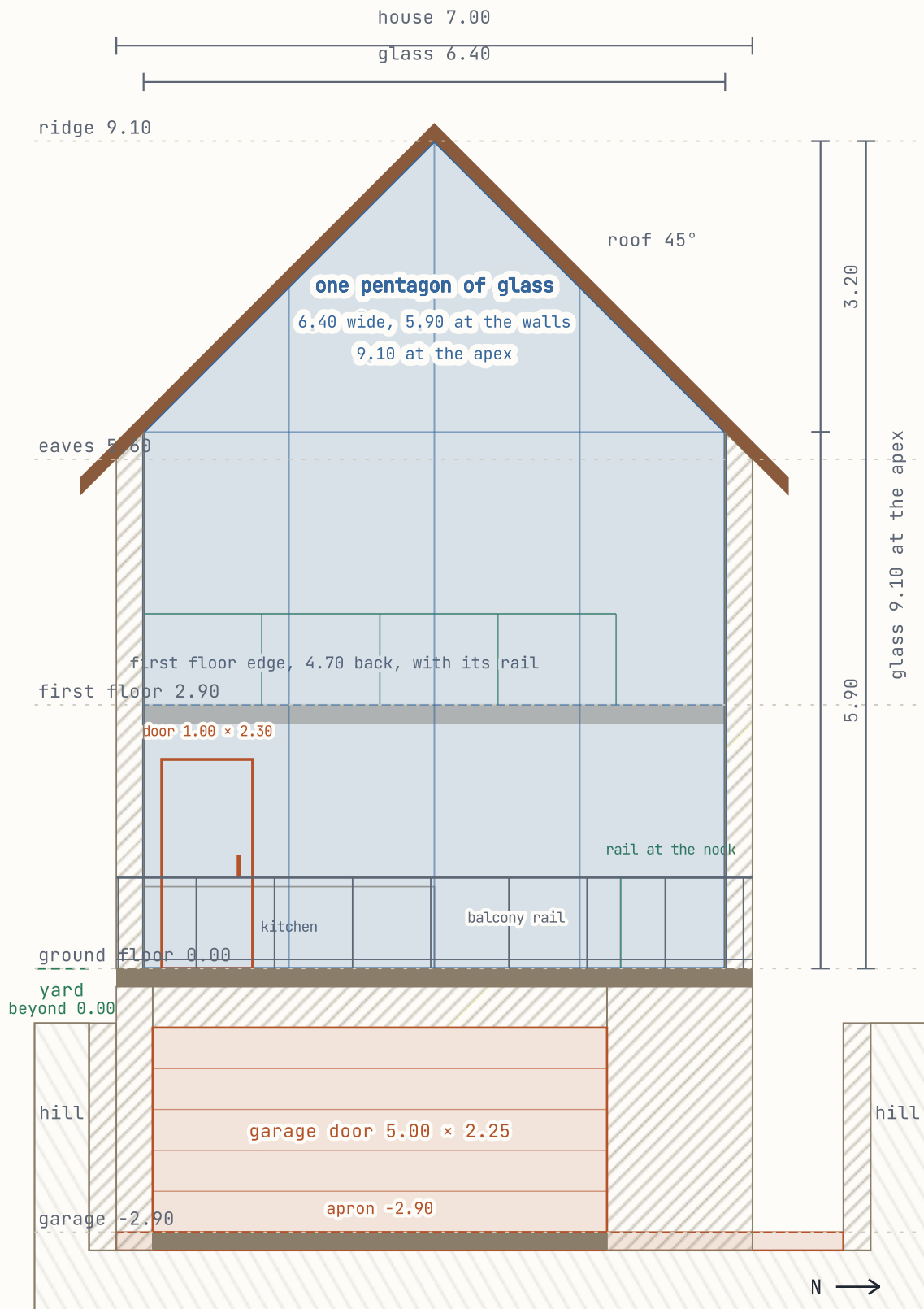


- **The top riser is the garage's back wall line, x 5.7:** the 17th step up lands on the ground floor over the garage's west wall, facing west with 2.30 m of landing before the stair wall and 3.40 before the back wall. That is the hall, in the floor's north-west corner, and the stair up leaves it at its far end, turning south.
- **The opening** runs from the 5th riser to the top, 3.48×1.20 along the north wall, in the ground floor's deck only: the first floor's deck is whole above it, 2.70 over the landing. The deck stays over treads 1 to 4, which is the nook by the glass.
- **Headroom is 2.00** over the 4th tread, the minimum. Starting the opening one riser earlier would give 0.17 more and leave a 1.23 nook.

- **The door between garage and house** belongs at the foot of the garage flight, on the 1.36 landing inside the garage door, swinging onto that landing; a door at the top would open onto the stair.
- **The stair up** mirrors this one: the same 17 risers on 0.27 treads, 4.32 of run, climbing south along the back wall from z 1.4 to z 5.72, through an opening in the first floor's deck from its 5th riser (z 2.48) with the same 2.00 over its 4th tread.

East face, from the valley: the pentagon

Looking west at the glass, north to the right as in the plan. The garage door under the balcony, the balcony at the living room's floor, the glass from there to the ridge — and only the living room behind it.



- **One pentagon of glass**, 6.40 wide between the north and south walls, 5.90 at the walls, 9.10 at the apex: about 47 m² of glazing, split by mullions every 1.60 and transoms at the first floor's level and the eaves line.

- **Seen through it:** the first floor's edge 4.70 back with its rail, the kitchen, and the rail of the garage opening at the nook.
- **The balcony door** is at the south end of the glass, 1.00×2.30 , where the balcony and the yard meet at the house corner.

Rooms

ROOM	CLEAR	M ²	LIGHT AND DOORS
Living room with kitchen	$4.00 \times 5.20 + 1.52 \times 1.20$	22.6	open to the roof, 5.90 to 9.10; the glass pentagon east with a 1.00 door to the balcony; yard door 0.90 and window 1.40 south; open to the hall
Bedroom	3.20×3.20	10.2	window 1.40 to the yard; door 0.90 from the lobby
Closet under the stair	1.00×3.20	3.2	from the bedroom; 1.2 high at its north end, 2.7 under the top landing
Hall: landing, stair foot and lobby	$2.30 \times 1.20 + 1.10 \times 1.10 + 2.00 \times 1.90$	7.8	the garage flight lands in it, the stair up leaves it; borrows light from the living room
Toilet	1.20×1.80	2.2	door 0.80 from the lobby; no window, mechanical vent; beside the stair
Stair up, its lower half	1.00×2.10	2.1	treads 1 to 8; nothing fits under them
Garage opening	3.48×1.20	4.2	the garage flight below, railed
Partitions		1.5	
Inside the outer walls	8.40×6.40	53.8	

Still open

- The first floor: 4.40×6.40 clear behind the tall room, less the stairwell, 24.9 m^2 — the stair arrives in its south-west corner, so the bathroom goes over the toilet beside it and the rest is bedroom; the attic over it is another room if the roof is kept at 45° .

- The back wall now stands a metre deeper in the hill (about +1.9 of ground against it), and the yard's west edge followed it to x 2: more retaining wall and more cut, for 6.4 m² a floor.
- The roof pitch: 45° drawn, which puts the apex at 9.10. At 35° it is 8.05 and the glass loses a metre.
- 47 m² of east glass: heat in winter and sun in summer want a plan — triple glazing, and shading or a deep roof overhang at the gable.
- The door at the foot of the garage flight: self-closing, fire-rated where the rules ask.
- A north window: the wall is 1 m from the boundary; the rules decide.
- Whether the outer walls sit inside the 9 × 7 (drawn) or outside it: outside would give 9.00 × 7.00 clear and push the east glass onto the balcony's first 0.30.